Target heart rates for the development of cardiorespiratory fitness

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ABSTRACT

SWAIN, D. P., K. S. ABERNATHY, C. S. SMITH, S. J. LEE, and S. A. BUNN. Target heart rates for the development of cardiorespiratory fitness. *Med. Sci. Sports Exerc.* Vol. 26, No. 1, pp. 112–116, 1994. The American College of Sports Medicine (ACSM) recommends the use of 40%, 60%, 80%, and 85% of maximal oxygen consumption ($\dot{V}O_{2max}$) as target values in developing exercise prescriptions. Further, the ACSM states that 55%, 70%, 85%, and 90% of maximal heart rate (HR_{max}) may be used as indices of these respective levels of $\%\dot{V}O_{2max}$ for the general population. The current study evaluated this relationship between $\%HR_{max}$ and $\%\dot{V}O_{2max}$ in apparently healthy, young adults. Eighty-one men and 81 women between the ages of 18 and 34 engaged in an incremental exercise test to exhaustion. Linear regressions of $\%HR_{max}$ and $\%\dot{V}O_{2max}$ were performed on each subject. From these regressions, target values of $\%HR_{max}$ were computed for each individual. Mean percentages of HR_{max} were 63%, 76%, 89%, and 92% at 40%, 60%, 80%, and 85% of $\dot{V}O_{2max}$, respectively. At all of these values of $\%\dot{V}O_{2max}$, the values obtained for $\%HR_{max}$ were significantly greater ($P < 0.001$) than those used by the ACSM. Fitness affected these results, particularly among men. High fit men averaged 2% higher in $\%HR_{max}$ than low fit men at any given value of $\%\dot{V}O_{2max}$.

HEART RATE, OXYGEN CONSUMPTION, FITNESS, EXERCISE PRESCRIPTION

The American College of Sports Medicine (ACSM) recommends that individuals obtain regular aerobic exercise for the development of cardiorespiratory fitness. If the individual is apparently healthy, the recommended intensity level is 60% to 80% of maximal oxygen consumption ($\dot{V}O_{2max}$), while the outside limits are given as 40–85% of $\dot{V}O_{2max}$ (1).

The ACSM suggests that one practical way of assessing the intensity level of exercise is to use a percentage of maximal heart rate (HR_{max}). According to the ACSM, 55% of HR_{max} corresponds to 40% of $\dot{V}O_{2max}$, 70% HR_{max} to 60% $\dot{V}O_{2max}$, 85% HR_{max} to 80% $\dot{V}O_{2max}$, and 90% HR_{max} to 85% $\dot{V}O_{2max}$ (1). These target values of $\%HR_{max}$ are based on a number of studies that determined regression equations for $\%HR_{max}$ vs $\%\dot{V}O_{2max}$ (3–8,10,11). For example, Hellerstein et al. (5) studied male cardiac patients and male normals and derived the equation: $\%\dot{V}O_{2max} = 1.41(\%HR_{max}) - 42.0$. Franklin et al. (4) studied the relationship in women, and combined the data of eight previous studies on men. Virtually identical regressions for the two genders were obtained, i.e.: $\%\dot{V}O_{2max} = 1.33(\%HR_{max}) - 37.3$.

If, for example, 60% and 80% of $\dot{V}O_{2max}$ are entered into the equations cited above, the resultant values for $\%HR_{max}$ are approximately 73% and 88%, respectively, which are close to the values used by the ACSM. However, this method of deriving values for $\%HR_{max}$ is somewhat flawed. The original studies (with the exception of ref. 6, a study of 14 cardiac patients) used $\%HR_{max}$ as the independent variable in linear regression analysis. Projecting a target value for $\%HR_{max}$ at a given $\%\dot{V}O_{2max}$ therefore requires transposing the equation. Transposition of a linear regression equation does not result in the same values as those that would be obtained if the regression had initially been performed with the dependent (or criterion) and independent (or predictor) variables reversed, i.e., a regression equation should not be used to predict the independent variable.

A separate problem, found in all of the previous studies, is that the data from many individuals were combined to produce a single regression for the group. Target values of $\%HR_{max}$ are then calculated from the one regression formula. It is more appropriate to perform a linear regression for each subject and use these individual regressions to calculate target values of $\%HR_{max}$. Then, mean values for the group can be determined.

Finally, only one of the previous investigations (4) examined the relationship in women. And that investigation was able to make only indirect comparisons to the previously determined relationship in men, as men and women have never been evaluated in the same study.

For these reasons, we felt that a new investigation of the $\%HR_{max}$ vs $\%\dot{V}O_{2max}$ relationship should be done, so

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that the recommended target heart rates could be directly measured, and so that a sex difference, if present, could be examined.

A secondary question in the study was to assess whether the fitness of the subjects influences the determination of these target heart rates.

METHODS

Subjects. Eighty-one men and 81 women volunteered for and completed the study. Written informed consent was obtained according to guidelines of the university institutional review board. Inclusion criteria for participation were: 1) between 18 and 34 yr of age, and 2) apparently healthy with no known cardiovascular risk factors (1). Subject characteristics are presented in Table 1.

Protocol. Height, weight, and skinfold thicknesses were measured, the latter to estimate % fat (9). Subjects were then prepped for lead II electrocardiographic monitoring using an on-line telemetry system.

Subjects were instructed in treadmill walking and then fitted with a Hans-Rudolph face mask for collection of expired gases. A graded exercise test using the Bruce protocol (2) commenced, and the subjects were not allowed to grasp the treadmill handrails, except as an indication to stop the test.

Heart rate was monitored continuously, and recorded over the last 10 s of each minute of the Bruce protocol, as well as during the final 10 s of exercise. Ventilation, oxygen consumption, carbon dioxide production, and the respiratory exchange ratio (RER) were calculated for each 20-s period of exercise on a Sensormedics 2900 metabolic cart and averaged over each minute of exercise. The metabolic cart was calibrated against known gases prior to each test. Subjects were encouraged to exercise to exhaustion, and the highest $\dot{V}O_2$ over any continuous minute of exercise was termed maximal $\dot{V}O_2$, provided RER equaled or exceeded 1.10.

Data analysis. Each 1-min value of HR and $\dot{V}O_2$ was expressed as a percentage of its maximum. Linear regressions were performed for each of the 162 individuals using paired data points from the end of each 3-min Bruce protocol stage and at maximum, with %$\dot{V}O_{2max}$ chosen as the independent variable. Using the individual linear regressions, the percentages of HR_{max} corresponding to 40%, 60%, 80%, and 85% of $\dot{V}O_{2max}$ were determined for each of the 162 subjects.

Summary statistics (mean ± standard error) were then calculated for the group of men and for the group of women. These values were compared to those used by the ACSM (i.e. 55%,70%,85% and 90%, respectively) using a Student *t*-test, with the ACSM value designated as a population norm.

To assess any possible influences of sex and of fitness, the men and women were both subdivided into high, medium, and low fitness groups of 27 subjects each, based on a ranking of their $\dot{V}O_{2max}$s. Values of %HR_{max} achieved by each subject at the four designated percentages of $\dot{V}O_{2max}$ were then compared among groups with a 2 × 3 × 4 ANOVA followed by Tukey *post-hoc* testing. For all statistical tests, significance was judged at an alpha level of 0.05.

TABLE 1. Physical characteristics of the subjects.

	Mean	SE	Range
Men			
Age (yr)	24	1	18–34
Height (cm)	181	1	165–206
Weight (kg)	81.8	1.3	59.5–115.0
Body fat (%)	14	1	2–33
Max RER	1.21	0.01	1.10–1.46
Treadmill time (min)	13.8	0.2	10.0–19.1
$\dot{V}O_{2max}$ ($ml \cdot min^{-1} \cdot kg^{-1}$)	49.9	0.9	33.2–68.7
HR_{max} (bpm)	191	1	160–210
Women			
Age (yr)	25	1	18–34
Height (cm)	163	1	135–180
Weight (kg)	58.4	0.8	42.3–81.6
Body fat (%)	23	1	12–36
Max RER	1.20	0.01	1.10–1.58
Treadmill time (min)	11.3	0.2	8.2–16.5
$\dot{V}O_{2max}$ ($ml \cdot min^{-1} \cdot kg^{-1}$)	40.1	0.8	28.9–63.1
HR_{max} (bpm)	194	1	170–207

RESULTS

A summary of the physiological characteristics of the subjects determined from their maximal exercise tests is presented in Table 1.

The mean (±SE) of the 81 linear regressions from the men was: %HR_{max} = (0.643 ± 0.010)%$\dot{V}O_{2max}$ + (36.8 ± 1.0), with r = 0.988 ± 0.001; and that for the women was: %HR_{max} = (0.628 ± 0.014)%$\dot{V}O_{2max}$ + (39.0 ± 1.3), with r = 0.977 ± 0.010.

The percentages of HR_{max} that were obtained by the subjects at designated percentages of $\dot{V}O_{2max}$ are illustrated in Figure 1. For both men and women, the %HR_{max} values were significantly ($P < 0.001$) greater than those in use by the ACSM at all levels of %$\dot{V}O_{2max}$. The discrepancy was approximately 8 percentage points at 40% of $\dot{V}O_{2max}$ (63% HR_{max} vs 55% HR_{max}), and fell to 2 percentage points at 85% of $\dot{V}O_{2max}$ (92% HR_{max} vs 90% HR_{max}).

The values of %HR_{max} for women averaged one percentage point higher than the men at each exercise intensity. However, the *F* ratio for a sex effect was not significant, though a trend was suggested as *P* was less than 0.10.

Mean (±SE) $\dot{V}O_{2max}$'s of the three fitness groups were: 59.2 ± 0.7, 49.7 ± 0.5, and 40.7 ± 0.6 $ml \cdot min^{-1} \cdot kg^{-1}$ for the men; and 47.9 ± 1.0, 39.4 ± 0.4, and 33.1 ± 0.5 $ml \cdot min^{-1} \cdot kg^{-1}$ for the women. The values for %HR_{max} achieved by these different groups from 40–85% of $\dot{V}O_{2max}$ are presented in Table 2. There was a significant *F* ratio for an overall fitness effect on the results. Specifically, the high fitness group among the men obtained a %HR_{max} that was approximately two

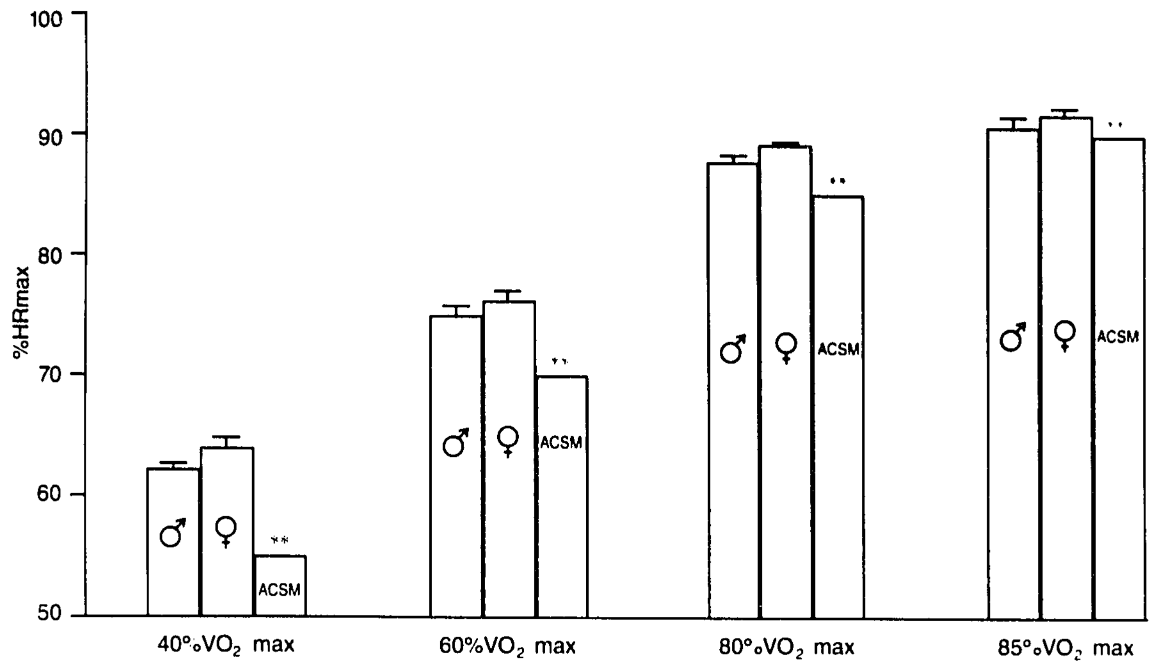

Figure 1—Percentages of HR_{max} attained by subjects at indicated percentages of $\dot{V}O_{2max}$. **Significant difference between ACSM standards and both men and women (P < 0.001).

TABLE 2. $\%HR_{max}$ at indicated $\%\dot{V}O_{2max}$s by fitness groups.

	40%	60%	80%	85%
Men				
Overall	62.5 ± 0.6	75.4 ± 0.5	88.2 ± 0.3	91.5 ± 0.3
High fit	63.7 ± 1.1**	76.4 ± 0.8**	89.1 ± 0.5*	92.3 ± 0.4
Medium fit	62.9 ± 1.0*	75.6 ± 0.7	88.3 ± 0.5	91.5 ± 0.4
Low fit	60.9 ± 1.2	74.1 ± 0.8	87.3 ± 0.6	90.6 ± 0.5
Women				
Overall	63.9 ± 0.8	76.4 ± 0.6	88.9 ± 0.5	91.9 ± 0.5
High fit	64.3 ± 1.1	76.6 ± 1.1	89.0 ± 1.3	92.1 ± 1.3
Medium fit	64.8 ± 1.2**	77.0 ± 0.8	89.7 ± 0.4	92.3 ± 0.3
Low fit	62.6 ± 1.4	75.7 ± 1.1	88.1 ± 0.5	91.3 ± 0.4

* Significantly different from low fit group (P < 0.05).
** Significantly different from low fit group (P < 0.01).

percentage points higher than the low fitness group at all levels of $\%\dot{V}O_{2max}$ studied. The effect among women was not as strong (Table 2), although the ANOVA did not reveal a sex × fitness interaction.

DISCUSSION

The principal finding of this study was that apparently healthy, young adults averaged 63% of maximal heart rate when at 40% of maximal oxygen consumption and averaged 92% of maximal heart rate when at 85% of maximal oxygen consumption. These heart rate values are significantly greater than the 55% and 90% recommended by the American College of Sports Medicine (1), and which are in widespread use for exercise prescription. The 8 percentage point discrepancy in $\%HR_{max}$ at 40% of $\dot{V}O_{2max}$ represents a 15% increase over the ACSM value (8/55), while the 2% discrepancy at 85% of $\dot{V}O_{2max}$ is only about a 2% increase (2/90). A smaller difference at higher intensities is expected, as the ACSM regression and the average regression from this study both approach a termination point of 100% HR_{max} at 100% $\dot{V}O_{2max}$. It must also be noted that the ACSM recommendations are intended for the entire adult population, while the current study is restricted to apparently healthy adults of 18–34 yr of age.

No previous studies have directly measured the values for $\%HR_{max}$ at specific intensities of $\%\dot{V}O_{2max}$. Earlier studies have determined linear regressions for $\%HR_{max}$ and $\%\dot{V}O_{2max}$ from which estimates of $\%HR_{max}$ have been made. As discussed earlier in this paper, these previous estimates may be inaccurate due to the mathematics involved.

The inaccuracy of these previous estimates does not reflect a poor design of the original studies so much as an inappropriate application of them. All but one (ref. 6) used $\%HR_{max}$ as the independent variable in regression analysis. This choice is questionable, as heart rates do not elicit whole body oxygen consumption, while oxygen consumption is clearly one factor determining the demand for heart rate. Further, if $\%HR_{max}$ is chosen as the independent variable, the resulting regression equation cannot be used to predict $\%HR_{max}$ at given levels of $\%\dot{V}O_{2max}$. We chose $\%\dot{V}O_{2max}$ as the independent variable, allowing us to predict target values of $\%HR_{max}$ for the purposes of exercise prescription.

An additional problem in all of the previous studies is that the data from many individuals were combined to produce a single group regression. This procedure eliminates individual variability from the results. The error involved is similar to the fact that the product of two means does not necessarily equal the mean of a set of products, i.e.: $(\Sigma a_i/n)(\Sigma b_i/n) \neq (\Sigma a_i b_i)/n$. To accurately reflect the variability within the data, the latter term should be used. We performed regressions for each subject, and used these regressions to determine each subject's %HR$_{max}$ values at the target %$\dot{V}O_{2max}$ intensities.

Among previous studies, only one examined the %HR$_{max}$ vs %$\dot{V}O_{2max}$ relationship in women (4), while several have done so in men (3, 5–8,10,11). This is the first study to examine the relationship in men and women using the same procedures and laboratory setup. There was no significant difference between the heart rate responses of women and men, although a trend ($P < 0.10$) for slightly higher (1%) values of %HR$_{max}$ among the women at any level of %$\dot{V}O_{2max}$ was observed. Even if confirmed in subsequent studies, such a small difference is inconsequential for the development of exercise prescriptions.

Differences in the %HR$_{max}$ obtained at given levels of %$\dot{V}O_{2max}$ were observed across fitness groups. To achieve a given percentage of $\dot{V}O_{2max}$, highly fit subjects had to attain a higher percentage of HR$_{max}$ than did the least fit individuals. Franklin et al. (4) previously showed that cardiovascular conditioning by a group of women caused a significant shift in their %HR$_{max}$ and %$\dot{V}O_{2max}$ regression. This shift resulted in a higher %HR$_{max}$ at a given %$\dot{V}O_{2max}$ in the more highly conditioned state, supporting the findings of the current study.

The influence of fitness can be explained on a mathematical basis. Since well-conditioned individuals have high $\dot{V}O_{2max}$s, their $\dot{V}O_2$ at a given submaximal workload represents a smaller percentage of $\dot{V}O_{2max}$ than it does for those who are less trained. For example, the first minute of the Bruce protocol elicited 20% of $\dot{V}O_{2max}$ from the highly fit men in this study, and elicited 30% of $\dot{V}O_{2max}$ from the least fit men (Fig. 2). This shifts the regression of the highly fit men to the left of the regression for the least fit men. The highly fit men also have a lower heart rate at this submaximal workload that partially compensates for the shift; however, the overall result is that the highly fit men attain a higher %HR$_{max}$ at any relative workload (%$\dot{V}O_{2max}$) compared to the least fit men.

Although the influence of fitness was statistically significant, it was quantitatively small. It may not be important enough to consider in the development of exercise prescriptions, except on an individual basis.

Limitations. As described in the methods, target values of %HR$_{max}$ were derived from linear regression analyses of each subject's incremental exercise data, as opposed to the alternative approach of attempting to steady state subjects at specified levels of %$\dot{V}O_{2max}$ and then measuring the resultant %HR$_{max}$ values. The use of regression analyses to interpolate the %HR$_{max}$ values introduces a source of error. However, every subject exhibited a correlation coefficient well above 0.9, with the average correlation coefficient being 0.988 for the male subjects and 0.977 for the female subjects (as reported in the results). Thus, the regression lines were extremely straight, providing highly accurate predictions of %HR$_{max}$. Based on these correlation coefficients and the typical n for each subject's regression and representative values for the standard error of y, the SEEs of individual regressions are estimated at 2 to 3%HR$_{max}$ units.

It is unlikely that the more cumbersome approach of attempting to have each subject steady state at the various prescribed levels of %$\dot{V}O_{2max}$ would result in more accurate heart rate data. For example, when the subject is intended to be at 40% of $\dot{V}O_{2max}$, he or she may actually be between 37% and 43% of $\dot{V}O_{2max}$, introducing a source of error. Even if each subject could be placed at precisely 40% of $\dot{V}O_{2max}$ (and then 60%, 80%, and 85%), the resultant "steady state" %HR$_{max}$ values would still carry an inherent source of error, i.e., each %HR$_{max}$ value would in fact be an average of several measure-

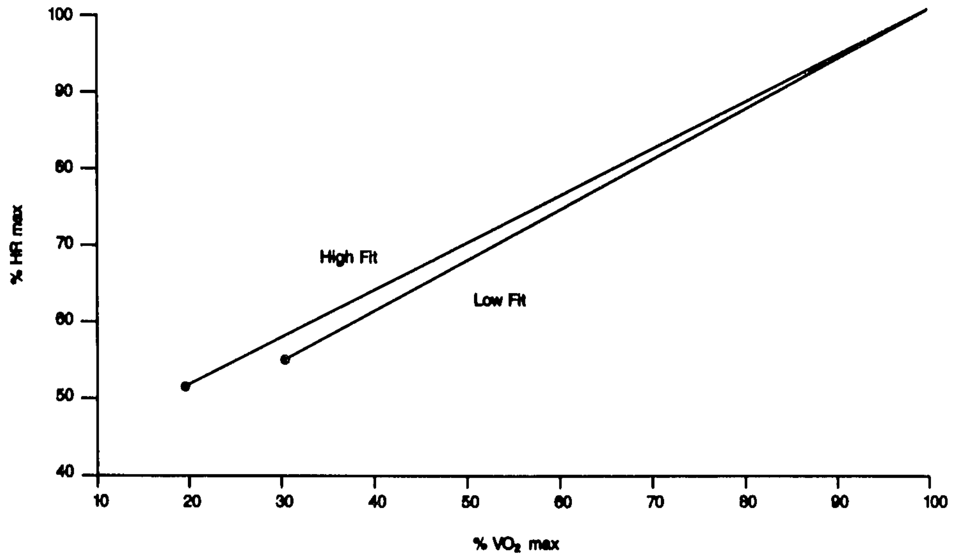

Figure 2—Average of 27 linear regressions in high fit males (%HR$_{max}$ = (0.636 ± 0.017)%$\dot{V}O_{2max}$ + (38.2 ± 1.7), with r = 0.987 ± 0.002), and of 27 linear regressions in low fit males (%HR$_{max}$ = (0.659 ± 0.017)%$\dot{V}O_{2max}$ + (34.6 ± 1.8), with r = 0.991 ± 0.002). The points of origin for the lines represent the average values attained during the first minute of the Bruce protocol.

ments and could more properly be considered as a mean and associated standard error. Thus, it is doubtful that such an approach would provide more accurate heart rate data than the regression method used in the current study.

A further question exists of whether heart rate values obtained in an incremental exercise test provide a true reflection of those obtained under constant workload conditions. This constraint was addressed in the current study by using data from the end of each 3-min Bruce protocol stage (and maximum), rather than data from each minute, to derive the regressions. While 3 min may be insufficient to establish a true "steady state", it provides a reasonable approximation that is frequently employed in the literature.

Thus, the methods of the current study contain inherent sources of error that must be considered in their interpretation. To the authors' knowledge, this is the first study reported in the literature to directly assess the target heart rates recommended for use by the American College of Sports Medicine.

Conclusions. The $\%HR_{max}$s attained by apparently healthy, young adults at 40%, 60%, 80%, and 85% of $\dot{V}O_{2max}$ were substantially higher than those currently recommended as target heart rates for the development of cardiorespiratory fitness. The use of the values obtained in this study, i.e. 63%, 76%, 89%, and 92% of HR_{max}, should be considered.

Populations with a low functional capacity, such as the elderly and cardiac patients, should be examined for a reevaluation of this relationship in each specific group. Among the young adults in this study, a fitness effect was observed that may be evident in other populations as well.

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